# Multiple Recovery Paths in a Lesioned Transformer

## Internal Rerouting and Readout Remapping

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BlackboxNLP 2026 | Poster #192

## Motivation

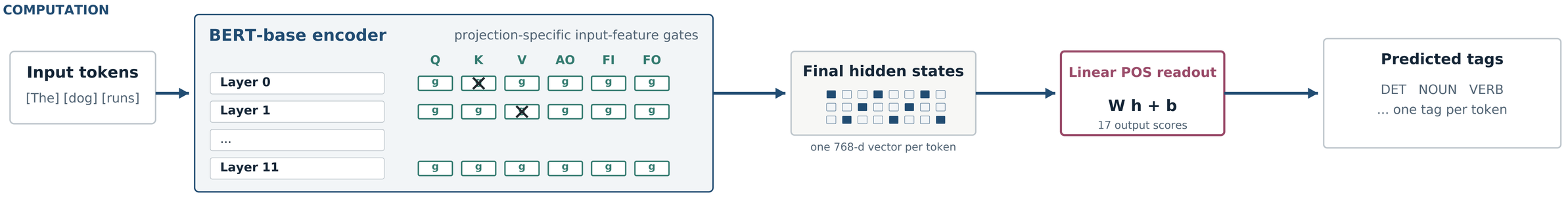
Using BERT as a frozen encoder and a linear part-of-speech (POS) readout as a task classifier, we first use gate masking to learn a sparse core subnetwork sufficient for POS classification. We then ablate this subnetwork to produce a large drop in performance. Our main question is whether performance can recover by adapting the surviving projection-specific gates, retraining the linear POS readout, or both.

**CENTRAL FINDING**

**Recovery can use internal rerouting or readout retraining; among fixed-weight routes, combining both recovers most without necessarily restoring the original representation.**

## Experimental framework

**COMPUTATION**




**EXPERIMENTAL PROGRESSION**

### A Learn the task-sufficient core

Train projection-specific hard-concrete gates and the linear POS readout while pretrained encoder weights remain fixed.

| | |
|---|---|
| Gates | TRAIN |
| Readout | TRAIN |
| Encoder weights | FIXED |

The sparse configuration reaches 94.98% primary-run POS accuracy.

### B Apply a permanent lesion

**Gates used by the task-sufficient core are permanently disabled. Their effective values are zero and gradients are blocked.**

74 Q/K/V 64-coordinate input groups
5 attention-output gate tensors
9 FFN-intermediate + 4 FFN-output tensors

**No recovery pathway can reactivate these entries.**

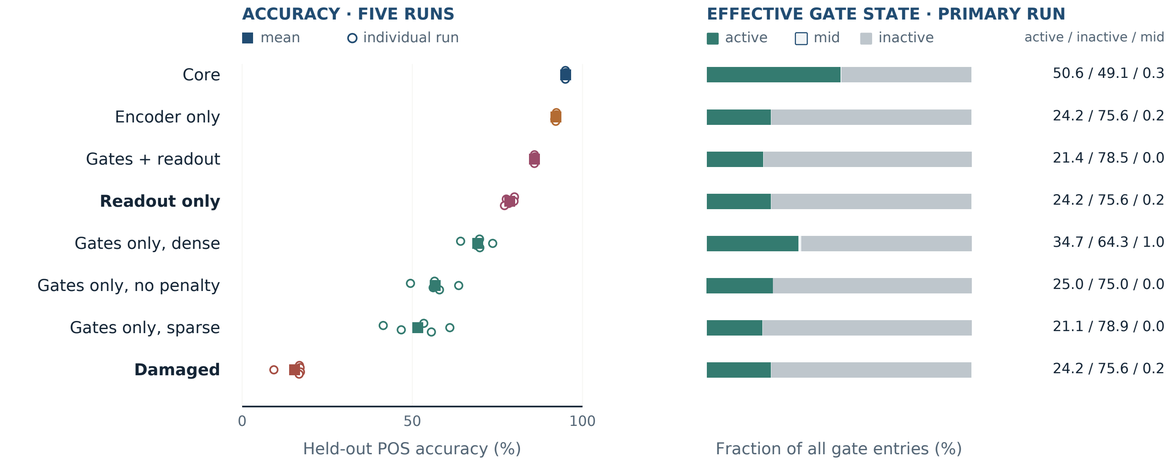
### C Recovery trainability

| Condition | Gates | Readout | Encoder | Lesion |
|---|---|---|---|---|
| Gates only | TRAIN | FIXED | FIXED | OFF |
| Readout only | FIXED | TRAIN | FIXED | OFF |
| Gates + readout | TRAIN | TRAIN | FIXED | OFF |
| Encoder only | FIXED | FIXED | TRAIN | OFF |

| | |
|---|---|
| SETUP | bert-base-uncased \| UD English-EWT \| 17 POS tags \| 25,147 development tokens |
| LESION | 74 grouped Q/K/V input-coordinate masks; 5 attention-output, 9 FFN-intermediate, and 4 FFN-output projection-mask tensors; 37,248 entries clamped to -12 and gradient-blocked |

## Performance and gate state

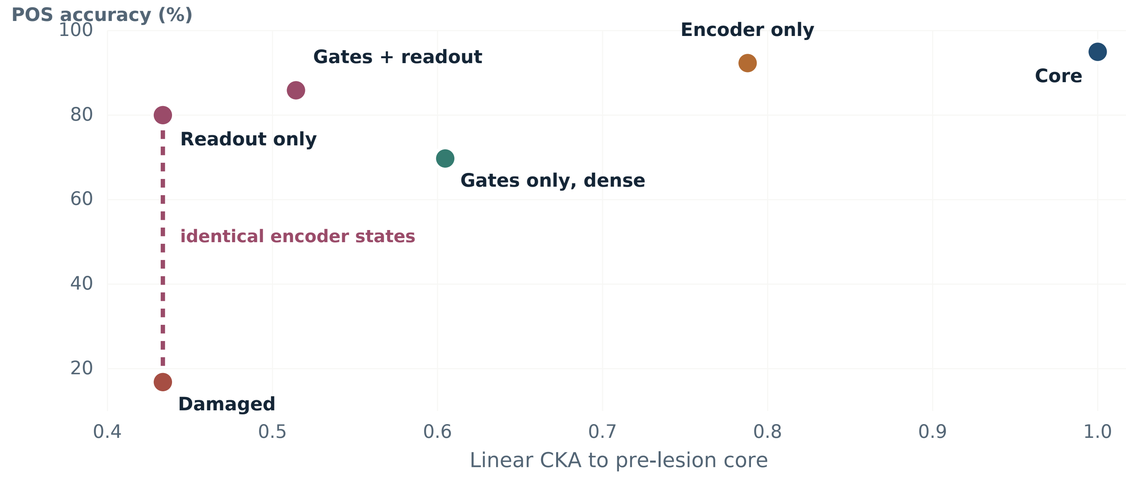
Five-run accuracy · primary-run gates




Circles are individual runs; squares are means. Gate state at the primary-run endpoint: active $p > 0.99$, inactive $p < 0.01$, otherwise intermediate. Lesioned entries are held at -12 and cannot change during recovery.

## Accuracy versus representation

Primary run

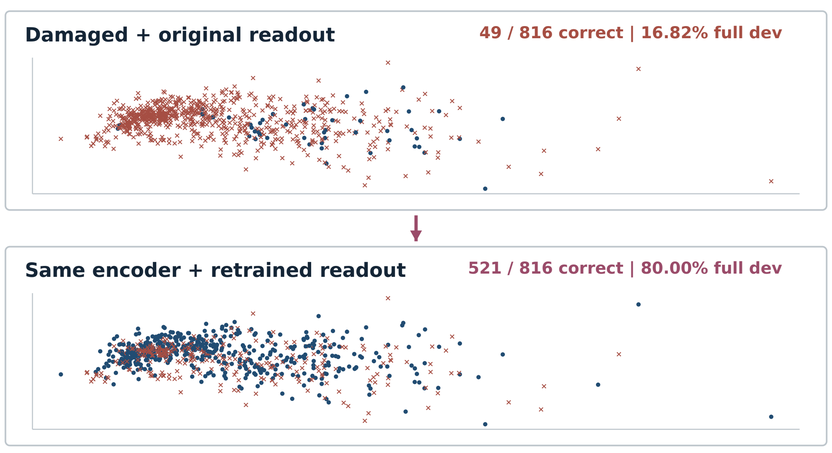



Linear CKA compares each condition's final hidden states with the pre-lesion core. Readout-only preserves the damaged hidden states (CKA 0.43) while primary-run accuracy rises from 16.82% to 80.00%.

## Encoder and readout geometry change differently across recovery paths

### Readout-only recovery

**Same token representations; only the linear readout changes**

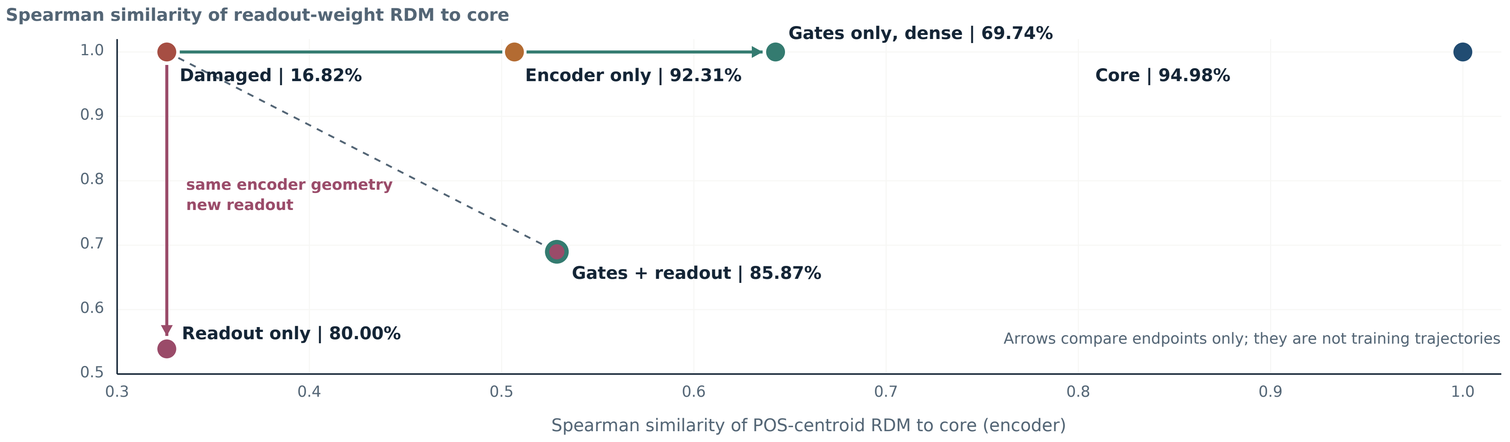



Coordinates are fixed across panels; only correctness under the original versus retrained readout changes. The shared core-fitted PC1-PC2 view explains 23.32% of core variance, while classification uses all 768 dimensions.

### Encoder and readout geometry

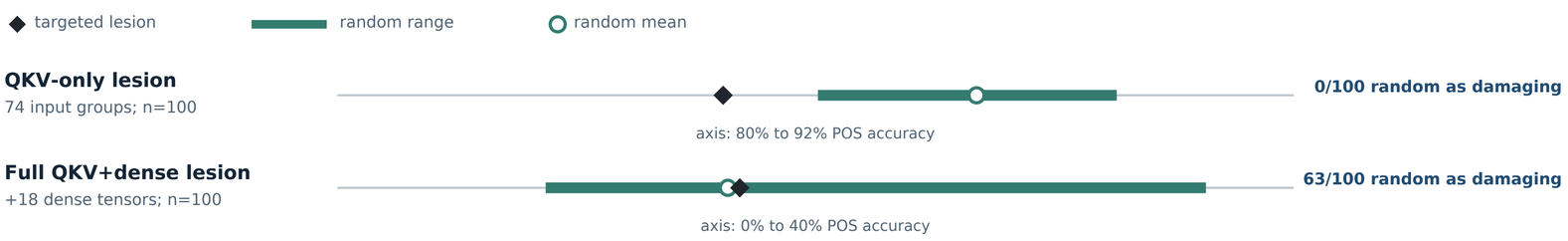
**Primary-run endpoints; labels report held-out POS accuracy**

**Spearman similarity of readout-weight RDM to core**




Readout-only holds encoder POS-centroid similarity at 0.33 while readout-weight similarity changes from 1.00 to 0.54 and accuracy reaches 80.0%. Gate-only and joint recovery also move the encoder geometry.

## Robustness and controls




Matched controls give different answers. The targeted Q/K/V-only lesion is more damaging than all 100 random 74-group lesions. For the full Q/K/V+dense lesion, 63/100 random matches are equally or more damaging; its severe endpoint is common under matched random targets.

## Conclusions

**Performance can recover without relearning the underlying transformations.**

The extent to which the original geometry is recovered depends on the degrees of freedom of the recovery protocol.

Circuit necessity is conditional on the permitted rerouting strategy. A component may be necessary under the original fixed routing and readout, but cease to be so when the readout can be retrained or surviving transformations can be reselected.

**LIMITATIONS**

**Our evaluations were based on a single task and a single backbone model.**

**EXPLORE THE EXPERIMENT**

**mechint25.github.io**

Methods: Louizos et al. (2018); Cao et al. (2021). Data: Universal Dependencies English-EWT.